

AXLE Planning

The product and plan specialist. Connects what the product needs with what demand requires, then explains what to make, what to buy and which dates or capacities are at risk.

Made for: product, engineering, operations and business teams.
Reading time: about 12 minutes.
Truth standard: describes the repository as implemented on 1 August 2026.

AXLE

What AXLE is here to do

A role, not an all-powerful chatbot

In one sentence

Connects what the product needs with what demand requires, then explains what to make, what to buy and which dates or capacities are at risk.

2

product areas grouped here

2

registered callable tools

3

allowed capability prefixes

0

agents it may delegate to

Receives

Customer dates from MICA · BOM and item definition · Stock/supply from SPAR · Capacity and execution state from KILN

Produces

Planned-order evidence · Date and capacity exceptions · Planning work item · A traceable explanation from demand to proposal

Allowed capability families

engineering.*

planning.*

agent.*

The key distinction

The product-area grouping below shows where AXLE contributes across XELOR. The callable-tool table later shows the smaller, exact surface Agent OS can invoke today. Product ownership does not automatically create runtime authority.

The business areas under AXLE

Business ownership is broader than today's Agent OS tools

Engineering

Item masters and bills of material—the product definition used by the rest of the plant.

Planning

Forecast, dated demand, MRP, planned orders, capacity, schedules and exceptions.

How to read this grouping

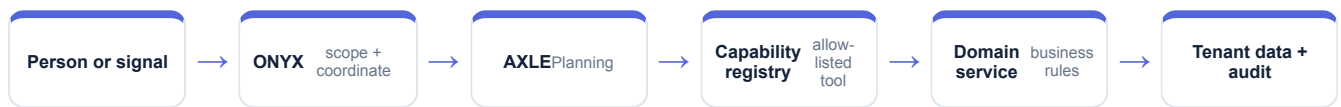
AXLE is the named specialist when a mission needs facts or analysis from these areas. Each module still owns its own rules, records and write paths. The agent coordinates through registered services; it does not become the database owner.

What remains true underneath

- Each module keeps its own business rules and permissions.
- Cross-module work passes through narrow ports or registered capabilities.
- Tenant isolation, audit and workflow apply regardless of which agent is speaking.
- A readable agent answer never replaces the source record.

How AXLE participates in a mission

The mission engine controls order, parallel work and recovery



1 · Scope

ONYX turns the goal into a versioned graph run with a fixed tenant, user, limits and expected outputs.

2 · Authorise

The capability registry checks that AXLE is allowed to call the named tool; RBAC repeats the user permission check.

3 · Execute safely

The registered domain service performs the operation. The agent never receives a general SQL doorway.

4 · Preserve evidence

Inputs, outputs, node events and checkpoints are stored so the mission can be explained or recovered.

Delegation

AXLE is a specialist and does not delegate to other agents in the current registry.

What AXLE can call today

Every row is explicitly registered and permission-checked

Capability key	Mode	Result	User permission	Approval?
planning.planned-orders.read	Read	MRP planned orders	planning.mrp.read	No
agent.action.dispatch	Execute	Governed work item	agentos.run.operate	Yes

What “dispatch” means

If listed, `agent.action.dispatch` appends a governed work item for a named specialist. It does not directly change a sales order, stock balance, production order, journal, payment or customer message.

Read / analyse / simulate

Returns evidence or a reproducible view. No business record is changed.

Draft

Creates reviewable proposed content. A draft is never labelled as an approved or completed business outcome.

Execute

Effectful at the Agent OS level and therefore requires an approved ancestor node.

Permission

The agent's allow-list and the signed-in user's RBAC must both allow the call.

Who AXLE works with

Specialists exchange evidence through ONYX, not hidden side conversations

MICA	Translates customer dates into demand without changing the promise.
SPAR	Separates buy needs from available material.
KILN	Shares make requirements and receives execution/capacity truth.

A complete controlled mission



Why this structure matters

Each agent stays accountable for its own facts. ONYX combines them, HEXA checks control evidence, and a person controls the decision boundary. This prevents one fluent answer from quietly becoming authority over the whole company.

What should we make or buy, and by when?

A realistic presentation path from question to controlled outcome

1

AXLE uses dated demand, forecast consumption, the multi-level BOM, stock, open supply, lead times and working-day calendars to produce planned orders.

2

Completed MRP workings are frozen so a later reviewer can see why a proposal existed. Exceptions expose impossible or late dates rather than hiding them.

3

Agent OS reads the resulting planned orders. Conversion into a real purchase or production order remains a controlled business step.

What the presenter should say

“AXLE contributes verified domain evidence, with source references and limitations. It does not claim that a proposed or dispatched action is complete until the controlled system records and verifies the outcome.”

Useful questions to ask during the demo

- Which source records support AXLE's answer?
- What permission was checked?
- Is this a fact, a simulation, a draft or a completed result?
- Would this step wait for human approval?
- What happens if a node times out or the service restarts?

How AXLE stays trustworthy

Controls are enforced in the runtime and modules, not left to prompt wording

Control 1

Completed MRP run details are immutable

Control 2

One planned order cannot be converted twice

Control 3

MRP planning and reorder-point planning cannot silently overlap

Control 4

Past-impossible release dates are surfaced, not backdated

Identity boundary

Verified user + tenant

Authority boundary

Agent allow-list + user RBAC

Action boundary

Approval ancestry for side effects

Data boundary

Domain service + tenant-fenced database

Evidence boundary

Append-only run, event and audit records

What is live—and what is not

This page prevents the demo from overclaiming

Live in the repository

- BOM-based planning foundation
- Planned-order read
- MRP and capacity logic
- Planning exceptions
- Demand-to-plan traceability

Deliberate current limits

- Agent OS exposes planned-order read, not every planning calculation
- No automatic order conversion
- Plan quality depends on dates, lead times, stock and BOM master data

Runtime disclosure

Orchestration, ERP reads, approval gates and governed action dispatch are live. Language reasoning is deterministic; no external model API or connector is active.

Primary implementation references

`apps/api/src/modules/engineering/`

`apps/api/src/modules/planning/`

`packages/platform/src/planning/`

This guide is a plain-language operating map, not an API contract. The code, migrations and automated tests remain the binding technical source.

AXLE in one page

Use this page when explaining the agent to a new team member

Job

Connects what the product needs with what demand requires, then explains what to make, what to buy and which dates or capacities are at risk.

Owns

Engineering, Planning

Receives

Customer dates from MICA · BOM and item definition · Stock/supply from SPAR · Capacity and execution state from KILN

Returns

Planned-order evidence · Date and capacity exceptions · Planning work item · A traceable explanation from demand to proposal

Can call

planning.planned-orders.read, agent.action.dispatch

Cannot do

Agent OS exposes planned-order read, not every planning calculation · No automatic order conversion · Plan quality depends on dates, lead times, stock and BOM master data

Plain-language glossary

Agent	A named specialist role with an allow-listed set of tools—not a free-roaming process.
Capability	One registered operation with an input, output, owner, mode and required permission.
Graph	A versioned mission recipe showing order, parallel steps, joins and approvals.
Checkpoint	A stored progress snapshot used for explanation and safe recovery.
Governed work item	An append-only assignment created after approval; it is not the final business mutation.